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Sourcing and designing with reused precast concrete elements: The ReCreate H22 pilot

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ABSTRACT: Reusing precast concrete elements from buildings condemned for demolition is a promising strategy for reducing the environmental impact of concrete in construction. While several successful cases of deconstruction and element reuse exist, scaling this approach from a niche practice to mainstream construction remains a significant challenge. One significant barrier is the conventional architectural design process, which is not well suited to integrating reclaimed structural elements into new projects. This paper addresses this challenge through the design and construction of a temporary exhibition building completed with 94% reused materials by weight. The project demonstrates a flexible design process that incorporates reclaimed precast concrete elements from multiple donor buildings into a new architectural configuration. The study emphasizes material sourcing as an integral aspect of the architectural design process and presents a systematic approach for identifying suitable donor buildings by collecting and analyzing demolition permits within a defined geographic area.

1 INTRODUCTION

1.1 *Reuse of precast concrete elements*

Currently, many concrete buildings are deemed obsolete and subsequently demolished, even though their technical service life is far from over. Salvaging concrete elements from buildings condemned for demolition, and reusing them in new load-bearing applications, is a promising strategy for reducing the environmental impact of concrete in construction. When effectively implemented, this approach can reduce demolition waste, decrease reliance on virgin raw materials, and lower the carbon footprint of new buildings. Given that the structural frame typically accounts for 60% of the total embodied carbon in multi-story buildings, reusing concrete elements presents a significant opportunity to reduce carbon emissions associated with new construction (Malmqvist et. al. 2023).

Circular Economy (CE) principles advocate for extending the use of products and materials for as long as possible at the highest possible value level. Accordingly, reusing buildings and components should always be prioritized over destructive demolition and downcycling. Precast concrete elements (PCEs) are a type of structural component that offers a substantial potential for circularity due to their modular design and durability (Huuhka et al. 2023). Unlike cast-in-place concrete structures, precast concrete buildings are conceived as modular systems that are prefabricated and assembled on-site. This construction method gained widespread adoption in the Post-war era, leading to a substantial proportion of Europe's building stock being composed of precast concrete elements (Alonso & Palmarola 2019). Although these buildings were not originally designed for disassembly, several projects have successfully demonstrated the deconstruction and reuse of precast concrete elements in new buildings. Since the 1980s, more than 50 completed projects have been documented utilizing salvaged precast concrete elements (Küpfer et al. 2023). However, while these projects demonstrate the

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technical feasibility of reusing precast concrete elements, the approach remains a niche practice and has yet to gain widespread adoption in mainstream construction (Huuhka et al. 2023). Several research initiatives are currently underway to address the technical, business, and regulatory challenges of scaling up concrete element reuse from isolated projects to the building sector at a system level. This study is carried out within the framework of the EU Horizon 2020-funded research project ReCreate: Reusing precast concrete for a circular economy. The project aims to advance knowledge and practices related to the circular use of precast concrete elements through four real-life pilot projects conducted in the participating countries: Finland, Sweden, Germany and the Netherlands (Huuhka et al. 2023).

1.2 *Design with reused building elements*

One of the challenges in integrating reused concrete elements into new projects is that the dimensions and properties of the salvaged elements dictate the feasible spatial configuration of the new building. In a conventional design process, the architectural concept of a building is typically developed before structural and material considerations, making it difficult to effectively incorporate a high proportion of reused structural elements into a project. In contrast, designing with reused precast concrete elements has been described as a "reverse" design process, where the design begins with a predefined stock of existing elements, rather than specifying materials after the completion of the design (Huuhka et al. 2023). Although the concept of a reverse design process may appear straightforward in theory, its practical implementation raises critical questions regarding its compatibility with the established phases and tasks in the building process.

Previous research has identified several ways in which designing and building with reclaimed building components is likely to differ from a conventional building project. Key differences include material sourcing and availability, the need for rigorous material assessment and quality assurance, the importance of early and close collaboration among involved stakeholders, and the need for a flexible design process and cost planning to accommodate material variability (Gorgolewski 2008, 2019, Addis 2009, Kozminska 2019). However, published accounts of completed projects incorporating reused precast concrete elements often provide limited insights into the planning and design processes, making it difficult to transfer strategies and solutions for reuse to future initiatives (Eklund et al. 2003, Asam 2007). Consequently, the role of architectural design in facilitating and maximizing the potential for concrete element reuse remains largely underexplored.

This paper seeks to expand the knowledge base on designing with precast concrete elements by documenting the design and construction of a real-life pilot project. The project involves the use of reclaimed concrete elements to construct a temporary exhibition building in Helsingborg, Sweden. Drawing on the authors' firsthand experience in leading the design process, the paper explores the challenges and potential of reusing precast concrete elements in architectural design. It specifically examines how the sourcing and availability of concrete elements became a critical design parameter, influencing not only the final building design but also project organization and the allocation of tasks and responsibilities throughout the project. Part two outlines the key phases in the project's development, emphasizing how reuse-related considerations influenced crucial design decisions. Part three presents the main findings and discusses the project's contributions to the growing body of knowledge on designing with reclaimed concrete elements.

2 THE H22-PILOT

2.1 *Project background*

Drottninghög is a mass housing area built on the outskirts of Helsingborg in southern Sweden. Originally developed in the late 1960s, it was conceived as part of a major national public housing program aimed at constructing one million housing units within a ten-year

period (Hall & Vidén 2006). Similar to many mass housing blocks from this period, the three-story buildings are composed of room-sized precast concrete elements. In the specific case of Drottninghög, the structural system consists of solid slabs and transverse load-bearing walls, commonly referred to as a "bookshelf" or "cross-wall structure". As part of an ongoing 20-year regeneration plan to improve area's socio-economic status, the city has decided to double the number of apartments while creating a more diverse building stock. As a result, some of the original buildings have been demolished, and more are under review in the planning process. Rather than merely discarding of the original buildings, the municipal housing company that owns them is interested in adopting a more circular approach. This strategy entails salvaging concrete elements from the existing housing blocks and repurposing them in newly planned residential developments within the area. Beyond the potential environmental benefits, the reuse of locally sourced materials could also contribute to promoting a more positive local history. To assess the feasibility of this approach, a small-scale pilot building was initiated within the framework of the ReCreate research project. The pilot had two primary objectives: to function as a temporary exhibition building for the Helsingborg H22 City Expo, and to generate data on various aspects of the deconstruction and reuse process for research purposes. From the outset, the project team included the municipal housing company, serving as both the client and project manager, alongside architects from the KTH Royal Institute of Technology, who were responsible for the architectural design and provided support on the reuse process.

2.2 *First donor building*

One of the original 1960s housing blocks in Drottninghög served as the initial donor building for the project. It was demolished in an early phase of the area's redevelopment, approximately one year prior to the official start of the pilot project. After consulting with the architects, the housing company decided to earmark and salvage eight elements (four wall elements and four slabs), which the demolition contractor set aside for use in the pilot project (Figure 1a). The elements were transported to a nearby waste management facility for intermediate storage (Vullings et al. 2024).

2.3 *Pre-design and second donor building*

In initial meetings between the client and architects, a suitable site for the temporary building was identified in Drottninghög, where an obsolete preschool was scheduled for deconstruction. The first concept sketches for the project were developed using the eight salvaged, room-sized concrete elements, complemented by additional materials sourced from the preschool. A key decision to influence the final design of the building was to reuse the existing raft foundation of the original school building (Figure 1b). This choice ultimately dictated the maximum footprint and the positioning of the new structure on the site. The boundary conditions for the new building were further defined by the existing detailed development plan, which set a maximum allowable building height of four meters. In addition to the raft foundation, doors and windows from the preschool were carefully salvaged during deconstruction for reuse in the upcoming pilot project.

2.4 *Schematic design, material sourcing method and third donor building*

Early in the design process, it was discovered that only five of the original eight salvaged elements were still available. In addition, improper handling at the storage site had caused considerable damage to the remaining elements. Upon inspection, it was determined that only one element was fully intact, while smaller fragments of the other four elements could still be salvaged. This unexpected change in material availability prompted the design team to begin searching for additional salvageable elements in the surrounding area. A method was developed to identify potential new donor buildings by collecting demolition permits from



Figure 1. Donor building 1: housing block, drottninghög (a); donor building 2: raft foundation from the preschool, drottninghög (b); donor building 3: industrial warehouse, helsingborg (c); and donor building 4: office building, lund, ultimately not used (d).

municipal planning offices in the region. In the initial screening, completed demolitions, as well as private properties and small-scale projects, were excluded from the search. The remaining buildings were then visually inspected using Street View in Google Maps. Based on noticeable indicators suggesting whether the buildings were precast concrete structures, the search could be further narrowed down. For the buildings deemed most promising, additional demolition permit documentation, which is publicly accessible under Swedish law, was subsequently requested from the relevant municipalities. To obtain a demolition permit, property owners are required to submit a demolition plan, material inventory, and recycling plan for the demolished material. Based on this material, buildings could be verified as concrete structures, narrowing the search down to 15 potential donor buildings. However, the available documentation did not always specify whether a concrete frame was prefabricated or cast-in-place, nor did it reveal the demolition timelines. Consequently, the design team reached out to property owners directly to confirm these details.

Ultimately, two buildings were identified as potential new donor buildings for the pilot project based on their (i) precast concrete frames, (ii) proximity to Drottninghög, and (iii) suitable demolition timelines. One was an industrial warehouse with a prefabricated concrete portal frame located only two kilometers from Drottninghög (Figure 1c), and the other was an office building in a nearby city featuring load-bearing exterior walls, columns, and hollow-core slabs (Figure 1d). When contacted by the design team, both property owners were positive about the prospect of salvaging elements for the pilot project. A preliminary design was developed to assess the architectural potential of reusing elements from the two buildings (Figure 2a). The new proposal was conceived as skeletal frame with two exhibition spaces, separated by an open passage. Shortened columns and beams from the warehouse (Donor Building 3) were repurposed for the main frame, while ten hollow-core slabs from the office building (Donor Building 4) formed the roof. The roof was proposed to be inclined to strengthen the connection between the interior and the exterior environment while enhancing the building's architectural expression (Figure 2a). Based on this preliminary design, the project team proceeded with securing the required elements from the respective donor buildings. With demolition work already underway at the warehouse site, the reuse intention had to be communicated quickly to the demolition contractor, and an agreement was made for the salvaged elements to be stored on-site after deconstruction, awaiting the start of the pilot construction. Obtaining

elements from the office building proved to be more challenging due to unforeseen planning delays, which ultimately postponed the demolition.

2.5 *Procurement of contractor, design development and building permit*

At this stage of the project, further development of the design was greatly aided by the procurement of a contractor. A partnering contract was established, allowing all stakeholders to collaborate closely to maximize the potential for reuse in the project. Through the contractor, the project team was expanded to include a structural engineer, a consulting architect for the planning application, and a subcontractor responsible for assembling the concrete frame on-site. This allowed for architectural design, structural engineering, and constructability issues to be addressed simultaneously, significantly expediting the decision-making process and ensuring that design details were developed to a sufficient level for the building permit application.

An initial challenge was the proposed sloped roof, which was ultimately replaced with a flat roof due to concerns regarding its complexity and safety during assembly. Another concern was the thickness of the raft foundation, which was discovered to be insufficient to support the concentrated point loads from the columns. This necessitated a specialized solution, leading to the decision to salvage the columns along with their pad foundations to facilitate a more even load distribution over a larger surface area. In addition to addressing a structural challenge, the foundation pads were also considered a valuable design feature, with the potential to integrate seating elements around each column. The demolition contractor was promptly informed of the new approach, allowing for adjustments in the deconstruction process to preserve the columns and their pad foundations intact. In contrast, the reused precast elements posed minimal structural concerns, as they were intended to span shorter distances and support lighter loads compared to their original application. Furthermore, an on-site visual inspection by the engineer confirmed that these elements were in satisfactory condition, reinforcing their suitability for reuse.

With the key structural elements of the building resolved, the design of the two exhibition rooms could be finalized, and the required documentation for the temporary building permit was prepared. One room was left open to display an installation made from the remaining salvaged elements from the original housing block (Donor Building 1), while the other room was enclosed, with one side covered in salvaged windows from the preschool (Donor Building 2) (Figure 2b). Ultimately, the dimensions of the ten hollow-core slabs defined the length and width of both rooms.

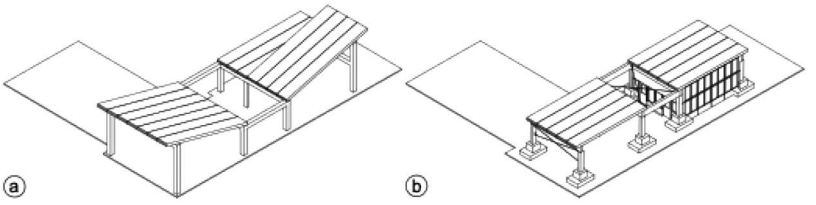


Figure 2. Early preliminary design of pilot project (a); the submitted design for building permit (b).

2.6 *Detailed design and construction documents*

While awaiting planning approval, it became evident that salvaging elements from the office building was unfeasible due to time and budget constraints, requiring an alternative solution for the roof. The subcontractor responsible for assembling the concrete frame, also one of Sweden's leading producers of precast concrete elements, offered the project a set of hollow core slabs that had been classified as "factory rejects" due to minor geometric discrepancies. However, only seven slabs were available at the time, which was resolved by introducing an open section above the outdoor exhibition room (Figure 3b). Meanwhile, the design team

finalized the remaining details for the construction drawings. The structural engineer developed special steel connections to ensure the safe and efficient assembly of the concrete frame. Bolt fixings were selected to minimize damage to the elements, facilitating disassembly and potential future reuse. An unusual solution was also devised to stabilize the column pad foundations on the existing raft foundation. Instead of permanent mortar, sand-filled boxes were used to accommodate irregularities on the underside of the footings. This sand base allowed for precise adjustments during assembly to achieve the desired alignment, while keeping the structure reversible for future disassembly.

2.7 *The completed building*

The pilot was completed in less than two months using 94% reused material by weight (Al-Najjar & Malmqvist 2025) (Figure 3a). The assembly of the concrete frame was completed in just three days, with the salvaged concrete elements delivered "just-in-time" from their respective storage sites. In addition to the concrete elements, reused materials included doors and windows from the preschool, as well as Oriented Strand Boards (OSB) sourced by the contractor from a construction site 50 km away. New materials included steel for the custom-made connectors, steel fasteners, tar paper for the roof, wood studs, and paint. Although "salvaged," the factory-rejected hollow-core slabs did not have a prior load-bearing function and are therefore also classified as new material in this study. Following the H22 City Expo, the pilot building was disassembled, and the precast concrete elements were transported to a municipal storage facility dedicated to the reuse of products and materials. The client intended to repurpose the hollow-core slabs for the construction of a parking garage in Drottninghög, but this project remained in the planning stage and did not progress. After a year in storage with no reuse opportunities, the elements were ultimately transported to a recycling facility, where they were crushed for recycling.

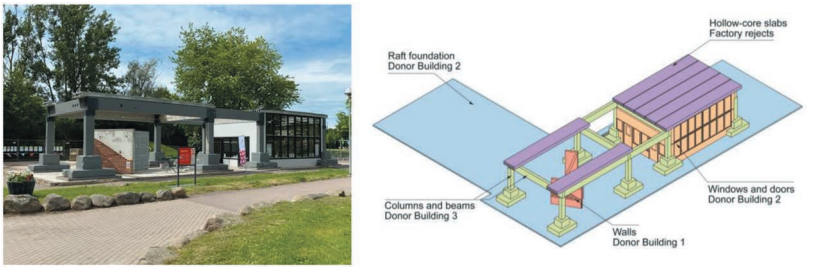


Figure 3. The completed building (a), and the reused building elements (b).

3 DISCUSSION

3.1 *An integrated design approach*

The high rate of reused building material in the pilot project was achieved through the dedicated collaboration of the entire project team, which included the municipal housing company, the project architects, the contractor, the structural engineer, and the subcontractor responsible for the on-site assembly of the concrete frame. The early involvement of all stakeholders was facilitated through a partnering contract, in which all actors committed to maximizing the use of reclaimed materials within the project's predefined budget. Engaging the contractor and engineer from the design development stage ensured that architectural and structural considerations were consistently evaluated alongside construction feasibility and cost. This integrated design approach facilitated efficient decision-making and led to innovative solutions that would not have been possible in a conventional, separated design process.

The use of reclaimed precast concrete elements as the primary building material required stakeholders to take on new responsibilities and roles throughout the project. For the engineer, this meant assessing the structural integrity of the reclaimed elements (Räsänen et al. 2024), a task not typically required in conventional projects. The client, in turn, had to assume ultimate responsibility for the quality of the elements due to the absence of established standards or CE marking, a responsibility typically handled by a concrete element manufacturer. For the architects, the reuse process involved actively engaging in the inventory and sourcing of concrete elements. By systematically securing demolition permits within a defined catchment area, they were able to identify two potential donor buildings for the project. After meeting the fundamental requirements of timely availability and geographic proximity, the reuse potential of the two donor buildings was ultimately assessed by examining their element stock and evaluating the suitability of these elements to meet the design requirements of the new building. While material sourcing is not traditionally part of an architect's responsibilities, their typical early involvement in the building process presents a valuable opportunity to facilitate concrete reuse in building projects. By identifying suitable reclaimed materials and advocating for their integration, architects can expand their role beyond conventional practice.

3.2 *A flexible design process*

Two main design approaches when building with reused components are typically discussed in the literature: (i) design from a given material stock, and (ii) sourcing material stock to fit a predetermined design (Gorgolewski 2008). The H22 pilot project exemplifies a hybrid of these two approaches, as the salvaged concrete elements became available at different stages of the project's development, with distinct implications for logistics and design (Figure 3).

Before the demolition of the Drottninghög housing block (Donor Building 1), the architects conducted an inventory to identify the most suitable elements to salvage for the upcoming pilot project. This process involved analyzing the original construction drawings and developing an in-depth understanding of the building system's specific characteristics. While this approach allowed for careful selection, the intermediate storage phase posed a risk, with most of the salvaged elements being damaged. In contrast, reusing the raft foundation (Donor Building 2) presented fewer logistical challenges. The preschool was deconstructed closer to the construction timeline, and the foundation was reused in its original location. However, a complication arose later in the design process when it was discovered that the structural capacity of the foundation was insufficient to support the concentrated load of the concrete columns. Although this issue was later resolved, it underscores the importance of considering the structural properties, alongside the architectural potential, when evaluating a component's suitability for reuse.

The concrete elements sourced from the industrial warehouse (Donor Building 3), had the greatest impact on the final design of the building. The warehouse, featuring a precast portal frame, introduced a new structural typology, shifting the available element stock from room-sized walls and slabs to large-scale columns and beams. This required a complete revision of the initial concept design, leading to the redesign of the pilot as a skeletal frame structure. The flexibility provided by the open design brief enabled the design team to implement this significant change at such an advanced stage of the project. The change in scale, from a large warehouse building to an exhibition pavilion, meant the columns and beams were excessively long for their intended use and required shortening. Ultimately, the oversized proportions of these components became a defining visual feature of the completed building. For several months, the design work proceeded under the assumption that hollow core slabs from the office building (Donor Building 4) would be available for the project. However, this posed a significant risk due to the uncertainty surrounding the procurement of the elements. Ultimately, the slabs were replaced with factory-rejected hollow core slabs of the same dimensions. This flexibility was made possible due to the highly standardized nature of hollow core slabs as a building component.

3.3 *Limitations of the study*

The conception and realization of the pilot project differed significantly from a conventional design and construction process using new building materials. As a temporary structure, the pilot offered both advantages and specific challenges for the reuse process. The temporary building permit meant that the pilot did not need to comply with the same regulatory framework required for permanent buildings, enabling the project to proceed on a simplified and accelerated timeline. Additionally, the temporary exhibition format allowed for an open architectural brief, offering more design freedom and flexibility in material choices, than is to be expected in a permanent structure. However, the fixed deadline imposed by the H22 City Expo also added additional pressure, making it more challenging to source reclaimed concrete elements from suitable donor buildings within the limited timeframe.

This paper has focused on the architectural design of the ReCreate H22 pilot. A more detailed account of the deconstruction of the donor buildings (Vullings et al. 2024), the quality assurance process (Räsänen et al. 2024), and the climate impact assessment of the completed pilot (Al-Najjar & Malmqvist 2025) has been published separately.

4 CONCLUSIONS

The ReCreate H22 pilot project was designed and completed using 94% reused material by weight, incorporating concrete elements from three different donor buildings. These reclaimed elements became available at various stages of the project, with distinct implications for logistics and design. In this way, the project exemplifies a hybrid model that combines two primary reuse strategies: (i) designing based on an existing stock of materials, and (ii) sourcing materials to fit a predetermined design.

Flexibility played a key role in achieving the project's high reuse rate in two main ways. First, a flexible design process allowed the project to adapt to changes in element availability while still meeting the design requirements of the new building. Second, the salvaged elements themselves offered a level of adaptability, as some were modified in both size and structural typology to fit their new function. This project underscores that matching donor buildings with new receiver buildings is likely to be a dynamic process, requiring a balance between these two approaches to maximize reuse potential.

For architects, early involvement in building projects presents a valuable opportunity to act as facilitators of reuse by advocating for the use of reclaimed materials from the initial design phase. A key contribution of this pilot project has been the validation of a method for identifying suitable donor buildings through the collection and analysis of demolition permits within a defined geographic area.

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